

Def. Sequence/List/Tuple

A sequence — also known as a list or tuple — is an ordered collection of objects, typically called components or entries. When the # of components in the collection is 2, 3, 4, ..., or n , the sequence is called an (ordered) pair, triple, quadruple or n -tuple, respectively.

— we write sequences, lists, tuples using angle brackets $\langle \dots \rangle$

Ex: $\langle 1, 2, 6 \rangle$ 3-tuple or triple

$\langle \text{Montana}, \text{Texas} \rangle$ 2-tuple or pair

$\langle 1, 2 \rangle \neq \langle 2, 1 \rangle$ | $\langle 0, 1, 2 \rangle \neq \langle 2, 0, 1 \rangle$

$\langle 1, \text{Texas}, \Delta \rangle$ ✓

Some people uses paranthesis to represent tuples

$\langle 1, 2 \rangle = (1, 2)$

$(1, 1) \leftarrow$ This is a possible pair

Def cartesian product of two sets

The cartesian product of two sets A and B is the set $A \times B = \{ \langle a, b \rangle : a \in A \wedge b \in B \}$ containing all ordered pairs where the first element comes from Set A and the second element comes from set B .

$$\langle 0, 2 \rangle \neq \langle 2, 0 \rangle$$

$$\langle 0, 2 \rangle = \langle 0, 2 \rangle$$

ex. $A = \{0, 1, 2\}$ $B = \{2, 3\}$

$$A \times B = \{ \langle 0, 2 \rangle, \langle 0, 3 \rangle, \langle 1, 2 \rangle, \langle 1, 3 \rangle, \langle 2, 2 \rangle, \langle 2, 3 \rangle \}$$

$$B \times A = \{ \langle 2, 0 \rangle, \langle 2, 1 \rangle, \langle 2, 2 \rangle, \langle 3, 0 \rangle, \langle 3, 1 \rangle, \langle 3, 2 \rangle \}$$

Ex. $\mathbb{R} \times \mathbb{R} = 2D \text{ plane} / \text{cartesian plane}$

$$\langle 0, 0 \rangle \in \mathbb{R} \times \mathbb{R}$$

$$\langle 1.11234, 2.012345 \rangle \in \mathbb{R} \times \mathbb{R}$$

$$\mathbb{R} \times \mathbb{R} = \mathbb{R}^2$$

$$\{\text{red, blue}\} \times \{1, 2, 3\} = \{ \langle \text{red}, 1 \rangle, \langle \text{red}, 2 \rangle, \langle \text{red}, 3 \rangle, \langle \text{blue}, 1 \rangle, \langle \text{blue}, 2 \rangle, \langle \text{blue}, 3 \rangle \}$$

$$\langle 1, \text{red} \rangle \notin \{\text{red, blue}\} \times \{1, 2, 3\}$$

$$\langle 1, \text{red} \rangle \in \{1, 2, 3\} \times \{\text{red, blue}\}$$

$$\mathbb{R}^3 = 3D \text{ space}$$

$$\{0, 1\} \times \{2, 1\} = \{ \langle 0, 2 \rangle, \langle 0, 1 \rangle, \langle 1, 2 \rangle, \langle 1, 1 \rangle \}$$

$$\{0, 1\} \times \{2, 1\} \times \{2, 0\} = \{ \langle 0, 2, 2 \rangle, \langle 0, 2, 0 \rangle, \langle 0, 1, 2 \rangle, \langle 0, 1, 0 \rangle, \langle 1, 2, 2 \rangle, \langle 1, 2, 0 \rangle, \langle 1, 1, 2 \rangle, \langle 1, 1, 0 \rangle \}$$

They are fundamentally equivalent but not exactly the same

$$\underbrace{(\{0, 1\} \times \{2, 1\})}_{\text{set}} \times \underbrace{\{2, 0\}}_{\text{set}} = \{ \langle \langle 0, 2 \rangle, 2 \rangle, \langle \langle 0, 2 \rangle, 1 \rangle, \dots \}$$

Question : What is the size of

$$|A \times B| = |A| \cdot |B|$$

8								
7								
6								
5				.				
4								
3								
2								
1								
	a	b	c	d	e	f	g	h

$\{a, b, c, d, e, f, g, h\} \times$

$\{1, 2, 3, 4, 5, 6, 7, 8\}$

$(e, 5) \in \{a, b, c, d, e, f, g, h\} \times$

$\{1, 2, 3, 4, 5, 6, 7, 8\}$

Def Binary Relations

A Binary Relation R on sets A, B is a subset $R \subseteq A \times B$

$$\langle x, y \rangle \in R \quad \text{as } x R y$$

$$\langle x, y \rangle \notin R \quad \text{as } x \not R y$$

Example: Suppose P is the set of all people.

Then let us define the binary relation R_1 on set P and p, R_1 is "_____ is (blood) related to _____"

$$R_1 \subseteq P \times P$$

$$R_1 = \{ \langle x, y \rangle : x \in P \wedge y \in P \wedge x R_1 y \}$$

$\langle \text{Serena Williams, Venus Williams} \rangle \in R,$
 $\langle \text{Venus Williams, Serena Williams} \rangle \in R,$
"Serena Williams is (blood) related to Venus Williams".

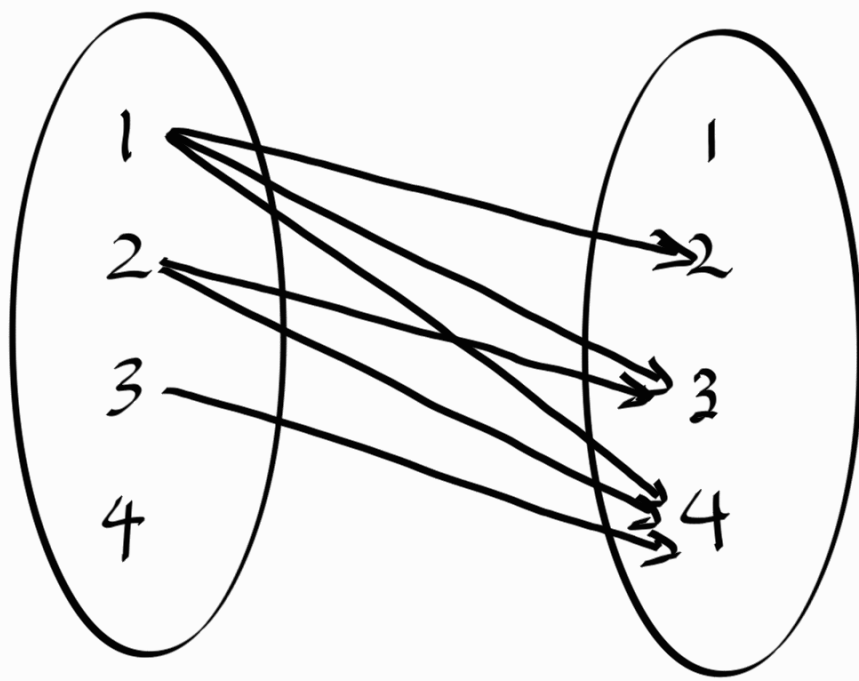
$\langle \text{Robin Williams, Serena Williams} \rangle \notin R,$

② Let $A = \{1, 2, 3, 4\}$, then we define $\langle$ relation as " is less than "

$$\langle \subseteq A \times A$$

$$\langle = \left\{ \langle 1, 2 \rangle, \langle 1, 3 \rangle, \langle 1, 4 \rangle, \langle 2, 3 \rangle, \langle 2, 4 \rangle, \langle 3, 4 \rangle \right\}$$

$$\langle \subseteq A \times A$$



③ Let $f: A \rightarrow B$ be a function.
Then let us define the set M
as follows.

$$M = \{ \langle a, f(a) \rangle : a \in A \}$$

is M a relation on sets A
and B ?

$$M \subseteq A \times B$$

for any function f , we can define a Binary Relation.

Let R be a binary relation on sets A, B

$$R \subseteq A \times B$$

$$R = \{ \langle x, y \rangle : x \in A \wedge y \in B \wedge x R y \}$$

Question: $f: A \rightarrow B$, where $f(x) = y$ is a function?

This is not true,

Look at < binary relation

$\langle 1, 2 \rangle, \langle 1, 3 \rangle$ exists.

$f(1)$ is not unique.

Ex! Let A be the set of months,
Let B be the days of a month.

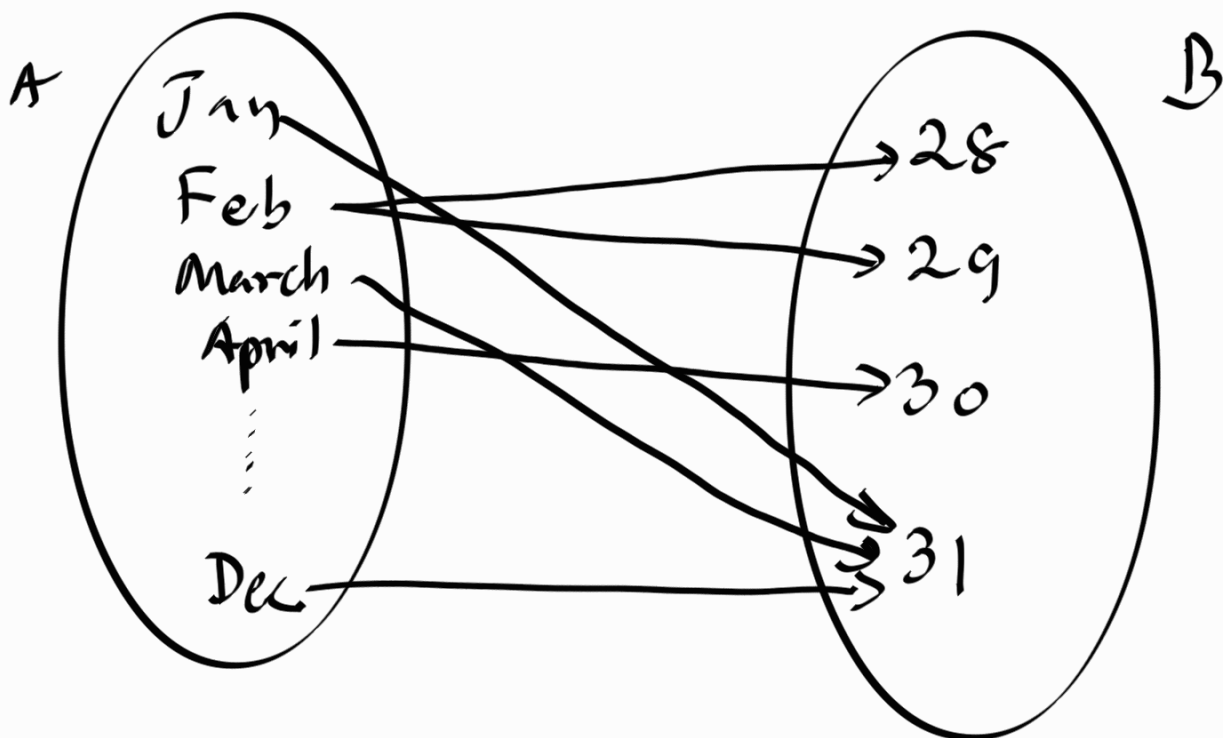
$A = \{ \text{Jan, Feb, March, April, } \dots, \text{Dec} \}$

$B = \{ 28, 29, 30, 31 \}$

R_1 : The pairs that contain the month
and its ^{max} # of days.

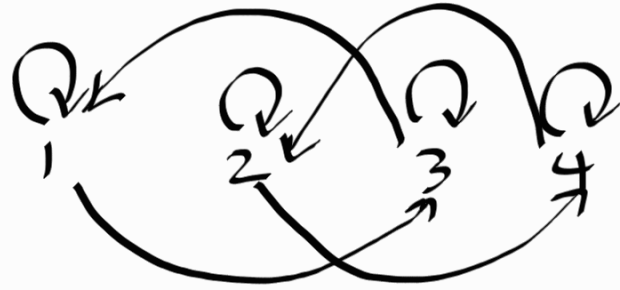
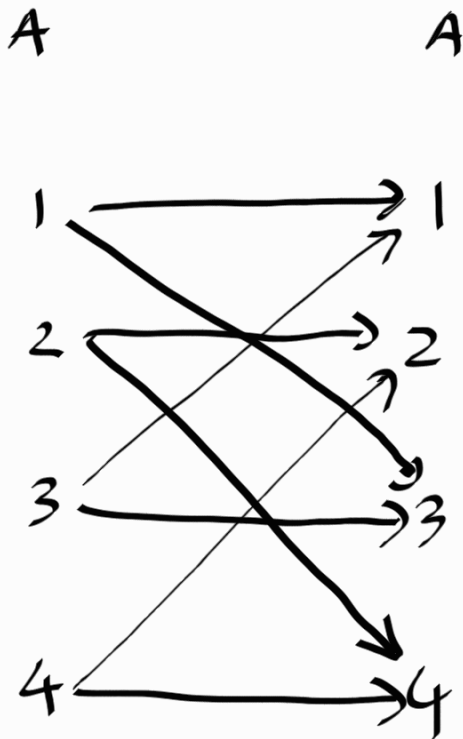
$$R_1 \subseteq A \times B$$

$R_1 = \{ \langle \text{Jan}, 31 \rangle, \langle \text{Feb}, 28 \rangle, \langle \text{Feb}, 29 \rangle, \langle \text{March}, 31 \rangle, \\ \langle \text{April}, 30 \rangle, \dots, \langle \text{Dec}, 31 \rangle \}$



$$A = \{1, 2, 3, 4\}$$

$$R_2 \subseteq A \times A$$



$$(1, 1) \in R_2 \quad (2, 4) \in R_2$$

$$(3, 2) \notin R_2$$

Relations on a single set

Let A be a set & let R be a subset defined on $A \times A$

$$R \subseteq A \times A$$

Properties of relations (when relation is defined on a single set)

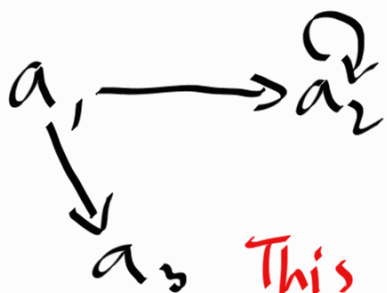
Suppose R is a relation defined on a single set A .

$$R \subseteq A \times A.$$

1. R is reflexive, iff $\forall a \in A: aRa$

Ex 1:

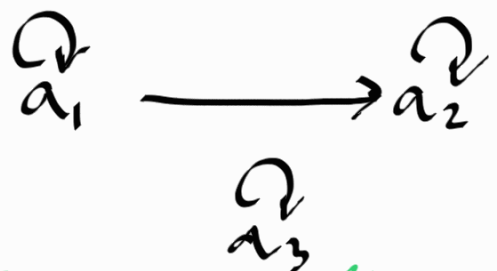
$$A = \{a_1, a_2, a_3\}$$



This is not reflexive

Ex 2:

$$A = \{a_1, a_2, a_3\}$$

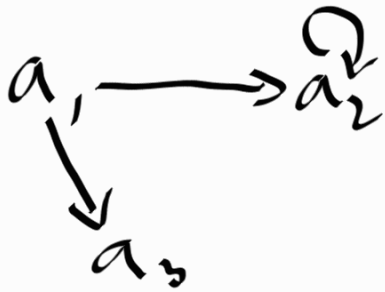


This is reflexive.

2. R is irreflexive iff $\forall a \in A: a \not R a$

Ex 1:

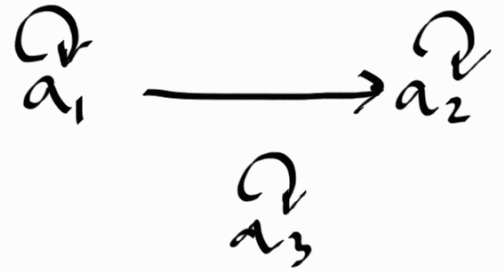
$$A = \{a_1, a_2, a_3\}$$



This is not irreflexive

Ex 2:

$$A = \{a_1, a_2, a_3\}$$



This is not irreflexive

Ex 3:

$$A = \{a_1, a_2, a_3, a_4\}$$

a_1 a_2

$a_3 \longrightarrow a_4$

This is irreflexive

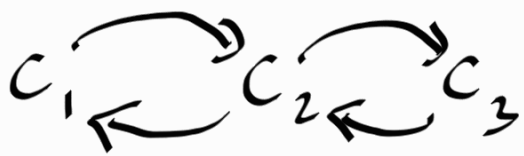
Ex 1: is neither reflexive nor irreflexive

3. R is symmetric iff

$$[\forall a_1, a_2 \in A : a_1 R a_2 \Rightarrow a_2 R a_1]$$

Ex 4:

$$C = \{c_1, c_2, c_3\}$$



Symmetric

Ex 5:

$$D = \{d_1, d_2, d_3, d_4\}$$



not - symmetric

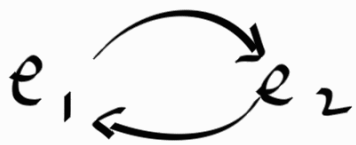
4. R is anti-symmetric iff

$$[\forall a_1, a_2 \in A : (a_1 R a_2) \wedge (a_2 R a_1) \Rightarrow a_1 = a_2]$$

In the graph, we can never have backward edges but self-loops are okay.

Ex 6:

$$E = \{e_1, e_2\}$$



Not anti-symmetric

Ex 7:

$$F = \{f_1, f_2\}$$

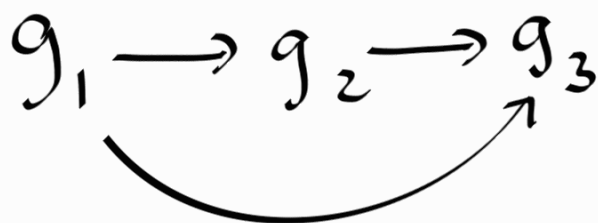


Anti-symmetric

5. R is transitive iff

$$[\forall a_1, a_2, a_3 \in A : (a_1 R a_2) \wedge (a_2 R a_3) \Rightarrow (a_1 R a_3)]$$

Ex 8: $G = \{g_1, g_2, g_3\}$

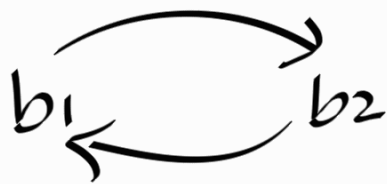


Transitive

$$a_1 = g_3 \quad a_2 = g_2 \quad a_3 = g_1$$

$$g_3 \not R g_2 \wedge g_2 \not R g_1$$

Ex 9:



$$B = \{b_1, b_2\}$$

$$a_1 = b_1 \quad a_2 = b_2 \quad a_3 = b_1$$

$$\overbrace{a_1 R a_2 \wedge (a_2 R a_3)}^{\text{I}} \\ (b_1 R b_2) \quad (b_2 R b_1)$$

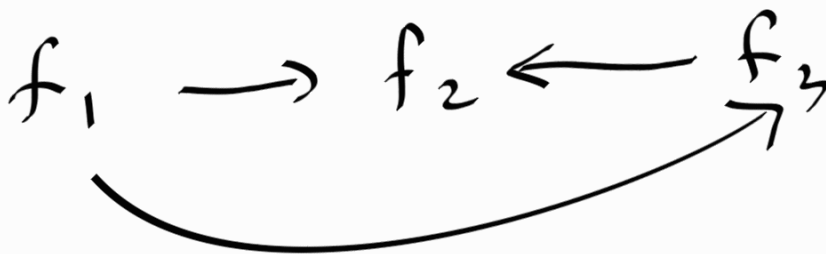
but I don't
have
 $a_1 R a_3$

$$(b_1 R b_1)$$

Not transitive

Ex 10:

$$f = \{f_1, f_2, f_3\}$$



Transitive

$$(f_1 R f_2) \wedge (f_2 R f_3) \Rightarrow$$

$$(f_1 R f_3) \wedge (f_3 R f_2) \Rightarrow f_1 R f_2$$

Ex 11:

$$G = \{g_1\}$$

$$\begin{array}{c} \circ \\ g_1 \end{array}$$

Transitive.

Suppose you have $<$ on $\mathbb{Z}$
 R_2 

$$R_2 \subseteq \mathbb{Z} \times \mathbb{Z}$$

$$(a, b) \in R_2 \text{ if } a < b$$

— is this reflexive? **No**

$$\forall a \in \mathbb{Z} : a \not< a$$

$$\forall a \in \mathbb{Z} : a \not R_2 a$$

or

$$\text{consider } a = 1$$

$$1 \not< 1$$

$$1 \not R_2 1$$

This is a counter example.

- is it irreflexive? **Yes**

WTS: $\forall a \in \mathbb{Z}: a \not R a$

$\forall a \in \mathbb{Z}: a \not\prec a$

because an integer is not less than itself.

$\forall a \in \mathbb{Z}: a \not R_2 a$

$\square$

- is it symmetric? **No**

Disproof by counter example

Let $a_1 = 1$ $a_2 = 2$

$a_1 < a_2$

by assumption

$a_1 R_2 a_2$

by def of R_2

$a_2 \not\prec a_1$

by assumption

$$a_2 R_2 a_1$$

by def of R_2

This violates the def of symmetric relation.

— is it anti-symmetric? yes

$$\text{WTS: } \forall a_1, a_2 \in A : (a_1 R_2 a_2 \wedge a_2 R_2 a_1) \Rightarrow a_1 = a_2$$

$$\text{Let } a_1, a_2 \in \mathbb{Z}$$

$$\text{Assume } (a_1 R_2 a_2) \wedge (a_2 R_2 a_1)$$

$$(a_1 < a_2) \wedge (a_2 < a_1) \quad \text{by assumption}$$

but this cannot happen.

Therefore, the premise of this implication is always false.

Hence, the implication is vacuously true.

- Is it transitive? **Yes**

WTS: $[\forall a_1, a_2, a_3 \in \mathbb{Z}: (a_1 R_2 a_2) \wedge (a_2 R_2 a_3) \Rightarrow (a_1 R_2 a_3)]$

Let $a_1, a_2, a_3 \in \mathbb{Z}$

Assume $(a_1 R_2 a_2) \wedge (a_2 R_2 a_3)$

$(a_1 < a_2) \wedge (a_2 < a_3)$

$a_1 < a_2 < a_3$

by assumption
and
by def R_2

$$a_1 < a_3$$

by def of
<

$$a_1 R_2 a_3$$

by def of
 R_2

R is transitive.